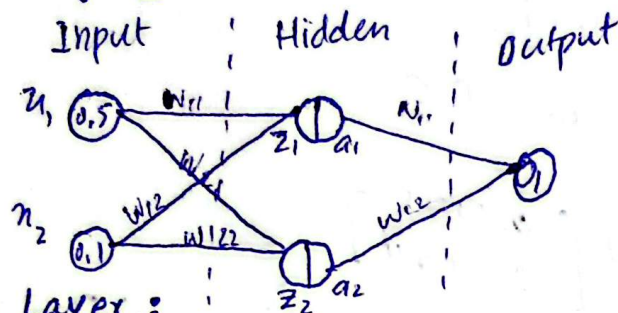


Task A : Forward Propagation

Input vector :  $x = \begin{bmatrix} 0.5 \\ 0.1 \end{bmatrix}$

True label :  $y = 1$



1). Hidden Layer :

$$z_1 = w_1 \cdot x + b$$

$$W_1 = \begin{bmatrix} 0.1 & 0.2 \\ 0.3 & 0.4 \end{bmatrix}, x = \begin{bmatrix} 0.5 \\ 0.1 \end{bmatrix}, b_1 = \begin{bmatrix} 0.0 \\ 0.0 \end{bmatrix}$$

$$\begin{aligned} W_1 \cdot x &= \begin{bmatrix} (0.1)(0.5) + (0.2)(0.1) \\ (0.3)(0.5) + (0.4)(0.1) \end{bmatrix} \\ &= \begin{bmatrix} 0.05 + 0.02 \\ 0.15 + 0.04 \end{bmatrix} \\ &= \begin{bmatrix} 0.07 \\ 0.19 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} z_1 &= W_1 \cdot x + b_1 \\ &= \begin{bmatrix} 0.07 \\ 0.19 \end{bmatrix} + \begin{bmatrix} 0.0 \\ 0.0 \end{bmatrix} \end{aligned}$$

$$\boxed{z_1 = \begin{bmatrix} 0.07 \\ 0.19 \end{bmatrix}}$$

2). Hidden Layer Activation :

$$a_1 = \sigma(z_1) = \frac{1}{1 + e^{-z_1}}$$

First element :

$$\sigma(0.07) = \frac{1}{1 + e^{-0.07}} \approx \frac{1}{1 + 0.93239} \approx 0.5175$$

Second element :

$$\sigma(0.19) = \frac{1}{1 + e^{-0.19}} \approx \frac{1}{1 + 0.82969} \approx 0.5474$$

Resulting vector :

$$a_1 = \begin{bmatrix} 0.5175 \\ 0.5474 \end{bmatrix}$$

3). <sup>Layer</sup> Output Linear Combination :

$$z_2 = w_2 a_1 + b_2$$

$$w_2 = [0.5 \ 0.6] , b_2 = 0.0$$

Dot product

$$w_2 \cdot a_1 = [0.5 \ 0.6] \begin{bmatrix} 0.5175 \\ 0.5474 \end{bmatrix}$$

$$= [(0.5)(0.5175) + (0.6)(0.5474)]$$

$$= 0.58719$$

$$z_2 = 0.58719 + 0.0$$

$$z_2 = 0.58719$$

4). Output Prediction :

$$\hat{y} = \sigma(z_2) = \frac{1}{1 + e^{-z_2}}$$

$$\hat{y} = \frac{1}{1 + e^{-0.58719}}$$

$$\hat{y} = 0.6427$$

5). Compute the Loss :

$$L = -y \log(\hat{y}) - (1-y) \log(1-\hat{y})$$

$$= -1 \log(0.6427) - (1-1) \log(1-0.6427)$$

$$= -1 \log(0.6427) - 0$$

$$= -(0.4421)$$

$$L = 0.4421$$